

MANUAL MALIN'S INTELLIGENCE SCALE FOR INDIAN CHILDREN

(Indian Adaptation of W.I.S.C.)

By
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CHAPTER 1

INTRODUCTION

Constructed after the general model of the popular American test of Dr. David Wechsler's Intelligene Scale for Children, better known by the acronym WISC, The Indian scale by Dr. Arthur J. Malin of Nagpur embraces all the advantages of the original along with what is hoped are several improvements.

Since its publication in 1949, the WISC has steadily gained in popularity over the older Stanford-Binet versions which dominated the American as well as the world scene for many years. A survey by Weiss amongst school psychologists in California in 1960 showed a strong preference for the Binet below the 3rd Grade but from the 3rd to the 12th grade the preferences were for the WISC.¹

Two main reasons for this preference for the WISC are the simplicity of its administration and its analytical breakdown into factorial functions for vocational and educational guidance. Unlike the several dozen unrelated test items of the Binet whose smooth manipulation requires long training,² the ten WISC subtests are repidly and easily learned.

Although Wechsler subscribes to some sort of "global concept" for the various intellectual capacities,³ his subtest with their separate scores for Verbal, Performance, Arithmetical, Manipulative, Abstract Reasoning, etc. abilities are a great boon for Guidance and Rehabilitation workers. With Global for Unitary IQ's being more and more supplanted by factor analytic test, this popularity of the WISC over the Binet is bound to wax and grow.

Although Terman's Binet versions have been introduced into India by such psychometric poineers as Rice and Kamat, in the respective versions of Hindustani Marathi and Kannada,⁴ the WISC makes its first introduction with this MISIC adaptation.

Adaptation - Good and Bad

Objections have been raised that adaptations of foreign tests cannot do full justice to the Indian child or testee.⁵ Such objections ignore the fact that psychological principles are free of cultural bias and as Dr. Mc Kenzie of Wilson's College Bombay states in the Forward of Dr. Kamat's test :

1. Weiss, American Journal of Mental Deficiency-Albany N. Y. Sept. 1962, No. 3181 - page 349.
2. V. V. Kamat Measuring Intelligence of Indian Children, III Edition, -Bombay : Oxford University Press, 1958 p.xii.
3. D. Wechsler, The Measurement and Appraisal of Adult Intelligence, - Baltimore : Williams and Wilkins Co., 1958 p.10.
4. A. Malin, Psychological Tests for Retarded Indian Children, Journal of Rehaibilitation in Asia. Bombay : Vol. IX No. 1, Jan. 1968.
5. K. G. Desai, The Construction and Standardization of a Battery of Group Tests of Intelligence in Gujarati, Pankore Naka, Ahmendabad : Bharat Prakashan, 1954, p. 34.

"There is nothing in the mental constitution of Indian children that warrants trying to devise tests radically different from those found suitable in the West."⁶

To close the doors to all foreign tests on the plea that indigenous constructions only can do full justice to the Indian environment would not only deprive us of the latest advances in psychometrics but also would scarcely commend India for a liberal scientific outlook.

We must of course rightly object to the misuse of foreign test in the so called "armchair" adaptations without any scientific standardization of samples and test items or the establishment of norms. Such adaptations can do more harm than good. And we regret to say that many Indian Psychiatrists and Teacher Training Colleges attempt to make such harmful use of intelligence tests. This misuse of tests also applies to tests adapted for Indian children but wrongly used for another language group without proper restandardization. A frequent case in point is to use the English translation of Dr. Kamat's Marathi and Kannada Binet when the test was never standardized or intended to be used for English speaking children.

How was the WISC Adaped

This brings us to the question of how the WISC had been adapted for Indian children in the MISIC. Wechsler seems to have been quite confident that transtated adaptations of his scale could be easily made. He wrote :

"The Scale has been translated into a number of languages. One of the things by which I have been most impressed is the relatively few changes that have been necessary."⁷

However in our case Wechsler would be quite shocked to learn that an almost total revision had to be made of the hopelessly culturally biased Verbal items of his scale even retaining the English without a translation.

Because of this radical revision which will be explained more in detail later, we have decided to call the adaptation by a new name viz., an Intelligence Scale for Indian Children -ISIC or MISIC.

Language Problem

Since English is the only language that applies universally throughout India at this stage, it was decided to adapt the WISC for the use of the English speaking child in India. On the other hand the 14 or more principal regional vernaculars were bound to acquire more and more dominance over English in their respective regions making regional vernacular adaptations of the WISC a necessity in the near future. The solution to this problem appeared to be a pilot culture free test in the English medium suitable for valid "armchair" translations or adaptions in the various regional vernaculars.

6. V. V. Kamat, op. cit. p.x.

7. D. Wechsler. op. cit. p. VIII

The Performance items naturally would not require translations but whether they would be sufficiently "culture free" for the Indian environment is open to discussion especially the "Spatial Relationship" factor of the Performance items.⁸

In our Indian adaption we have dropped the original Picture Arrangement test as too culturally biased but its use can be tolerated in the lower age levels for children who have been subjected to westernized educational and reading biases.

CHAPTER II

General Description of the WISC and MISIC

Before entering into the details of the major revisions involved in the adaption of the WISC it will be useful to give a general description of the test.

The original WISC is an individual intelligence test or scale for children from the ages of 5 to 15.11. The Indian adaption covers only ten years from 6 to 15.11.

The original Scale comprises twelve subtests divided into Verbal and Performance groups as follows :

Verbal	Performance
1. Information	1. Picture Completion
2. Comprehension	8. Picture Arrangement
3. Arithmetic	9. Block Design
4. Similarities	10. Object Assembly
5. Vocabulary	11. Coding
6. Digit Span	12. Mazes

The Indian adaption omits the Picture Arrangement (8) of the Performance Scale as it proved too culturally biased both as to content as well as to form.

These subtests may be administered in any order convenient for rapport. Only ten tests-5 from each group-are required for completed scoring. In case more or fewer tests are taken appropriate score pro-rating is called for. For safe validity however not less than four sub tests of each group should be taken. The Verbal group has an alternate in the Digit Span test in case of spoilage or when the regional vernacular test drops the vocabulary test in the upper levels (see paragraph on vocabulary Test Rationale). The Alternate should not be taken just to improve a score.

8. A. Malin, Indian Adaptation of Wechsler's Intelligence Scale for Children, Doctoral Thesis Unpublished, Nagpur University, 1966 p. 55.

Point Scale

The original WISC as well as its Indian adaption works on the Point Scale and all items of a given type are grouped together and arranged in increasing order of difficulty. Thus for example, all information items are in the subtest for "INFORMATION"

The points or raw scores of each test are totaled and then converted on the principle of the "Deviation IQ" into derived scores. In the original WISC these derived scores are standard scores called "Scaled Scores" which in turn must be converted into IQ's by means of a Table each for Verbal, Performance and Total Full Scale IQ's

The adapted MISIC for reasons to be explained later, avoids/ the use of the "Scaled Score" and by means of Tables converts the raw scores directly into "Test Quotients" which are actually IQ's. The Subtest TQ's are then added and group averaged (Verbal, Performance) and the Total or Full Scale of both groups is similarly obtained without the use of the Table.

Abandoning Mental Age Norms

Wechsler made much over his abandonment of the Mental Age norms in favour of his "Scaled Score" Deviation IQ technique. True, a Global M.A. suffers the same disadvantages of a Global IQ but Wechsler seems not to realize that his criticism levelled against the M.A. can be equally turned against the IQ's he uses in his test. He fails to distinguish between a global M.A. and a specific function or factorial M.A. This latter M. A. is very useful for education and guidance purposes. Moreover with all his criticism of the M.A. one wonders why he substituted another and even at that a meaningless "Scaled Score" intermediary between his raw scores and the final IQ's.

Vocabulary Test Rationale

Vocabulary tests in intelligence scales perform a dual role-one as a test of "Verbal Comprehension" and another as a test of "Verbal Information". This dual role however is never given a specific discriminating analysis. Most of the time as Wechsler explains :

"Only a knowledge of the number of words and their minimum content" is expected from this test.⁹ Correlation tables moreover show that in the upper age levels (10+) there is a very close correlation between Information and Vocabulary tests and the memory factor of the Digit Span test, but not so on the lower age levels (5-10)¹⁰

9. D. Wechsler op. cit. p. 126.

10. D. Wechsler, Manual Wechsler Intelligence Scale for Children, Psychological Corp. N. Y. 1949, p. 10-11.

From various other data it is clear that the Verbal Comprehension factor of the Vocabulary Test operates on the lower age levels and the Verbal Information factor on the upper age levels. The question then poses itself whether a Vocabulary test is actually needed at the upper age levels if the Information test sufficiently assesses that factor.

For obvious reasons then the Indian adaptation has dropped the Vocabulary test for the ages 10 to 16 in regional vernaculars and offers the Digit Span alternate to complete the 5th test required for the scoring. It should be noted however when administering the Vocabulary test in the regional languages that the scoring criteria for the lower age levels is as follows.

1. Definitions in "terms of use" or functional levels, e.g. knife cuts; are scored 1 point.
2. Definitions in "terms superior to use"-or concrete and descriptive levels, e.g. knife has sharp edge; are scored 2 points.

CHAPTER III

Standardization Statistics

Sample size

The original WISC was standardized on a sample of 100 "white" American boys and 100 girls at each age level from five through fifteen years. Each child was tested within one and a half months of his mid-year. Thus there was a total of about 2200 children. These samples were drawn from a varied geographical (urban-rural) school going population.

For the Indian adaptation over 1200 children were given full individual tests during the past 6 years and over 3000 were sampled in subtest trail runs. An average of about 90 samples were used for each age level including boys and girls in a 20-30 ratio. Age norms were based on a twelve month interval 4 months before and seven months after a birthday. Closer approximations are not possible due to the unreliability of Indian birth data. Comparisons with WISC birth data were made on the middle third of the WISC tables e.g. 6 yrs. 4 months through 6 yrs. 7 months.

Regional Norms :

Fortunately for English speaking children regional differences are at a minimum since the English medium schools follow fairly uniform standards in text books and literature content throughout the length breadth of India. The English version samplings were taken in Nagpur, Bombay, Simla, Manglore and New Delhi. The samples from the Hill boarding schools covered generally the whole northern portion of India.

For the Marathi and Hindi version samples were drawn from Nagpur alone which due to its centrally located position provides an ideal meeting place for the two above mentioned vernaculars.

Cultural Norms :

The original WISC as we noted above was standardized on a school going population in a country where school attendance is effectively compulsory. We realize with Dr. Bhatia that :

"the most characteristic feature of Indian life is the existence side by side of clearly mark groups-the one which has the benefit of literacy and formal education and the other which is deprived of it. Proportionately the illiterate group is very much larger."¹¹

Despite compulsory school legislation in India only a select few are able to find their way to or to remain in school. As menzel devastatingly remarked over 35 years ago-and recent statistics of school wastage still confirm his observation that :

"The Primary school is the end of school for the overwhelming majority" and whereas" the average age of pupils in class V is 12 years-the average Indian pupil of 12 years is not in Class V he is no longer in school ").. (!) ¹²

"The highly selective character of the Indian school population unavoidably "contaminates." our school samplings for the construction of IQ tests and produces serious normative problems about which we shall speak later, and raises the relevant question about the limited utility of IQ tests based on such biased populations.

Despite this high selectivity of the general school population in India we find the strange anomaly of stagnation in the two final age year groups of 14+ and 16+ . Our samplings show that these two years show lower IQ scores than the lower age levels. Evidently the samplings from these two years should be taken from the College population rather from the schools. Unless we make proprata scoring norms for these last two years we can highly consider the scores of the previous 8 year levels as valid until further research covers the College populations. The average American College entrance age is 17 whereas in India we find it closer to 14 and 15.

Varnacular Versions

Along with the English version two regional vernacular versions have been constructed, viz., Marathi and Hindi. Both of these versions have only about 300 samplings each. The Marathi samplings however have been consistently comparable to the English samples and have been joined to the same on an experimental basis in the construction of our norms. The Hindi samplings on the other hand for reason not yet known have shown disappointingly lower scoring results. We feel however for the time being until further researches confirm a definite inequality with the English and Marathi norms the Hindi scores can and should be interpreted against the composite norms of English and Marathi media.

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11. C. M. Bhatia, Performance Tests of Intelligence, Bombay : Oxford University Press, 1955 p. 2.
12. E. W. Menzel, Teaching, Bombay : March, 1939 p. 101.

Coefficients of Reliability and Validity

The original WISC reliability subtestwise by the Split-half method with appropriate correction for full length of the test by the Spearman-Brown formula and yielded a total coefficient of .91.

Our Indian adaptation established its reliability with the Test-Retest method and yielded a Pearson's Product moment correlation coefficient of .91 for the Full scale IQ results.

For the original WISC Wechsler does not provide any validity data and instead argues for some sort of "Construct Validity". Our Indian adaptation however has established Concurrent as well as Congruent Validity. The former was obtained from school ranking and came to .78 whereas Congruent validity was obtained from an adapted version of the California Short form Test of Mental Maturity for the upper age levels and from the Goodenough Draw-a Man test for the lower age levels. Both yielded a coefficient of .63.

CHAPTER IV

Establishment of Statistical Norms :

Wechsler's variant of standard scores which he called "Scaled Scores" were arrived at through a cumulative frequency distribution of raw scores for each test at each age level the percentile points being given standard score values on a theoretical normal curve with a smoothed progression. Our Indian norms were similarly arrived at, but our percentile points were directly converted into IQ's according to a modified Thomson formula also used in Dr. Bhatia's Performance scale.¹³

According to this method on a basis of a standard deviation of 15 IQ, the 5th percentile is equal to 75 IQ, the 16th to 85, the 50th 100, the 84th to 115 and the 95th to 125. From These anchor IQ's placed on a graph the intervening raw scores are normalized into respective IQ's. Because of the unusual platykurtic nature of our growth curve skewed to the right we have had to make two modifications in this Thomson formula.

The first modification was to add a hypothetical 5% to our lower second and third standard deviations to make up for the lack of subnormal cases since all our samplings were normal children and highly selective at that as we saw above. The second modification was at the other extreme.

The characteristic Indian psychometric phenomenon of high and widely scattered scores at the upper tail of the distribution makes the 95th percentile and abnormally high scoring norm when compared to the American WISC. Since the concentration of Indian scores lies mainly between the 16th and 84th percentile anchors we have normalized the upper second and third standard deviation IQ's by a line drawn from the average direction of the graphs between the 16th and 84th percentile anchor points.

13. C. M. Bhatia, op. cit. p. 62.

Verbal and Performance Score Disparities

Dispite the adjustments made to counteract the abnormal distribution of Indian scores we are confronted with two serious disparities in score results. In the first place a wide disparity exists between the American Verbal and Performane norms when compared to our present Indian norms and in the second place the Indian Verbal and Performance scores show a wider disparity between themselves than do the same two groups in the American version.

These two disparities are caused on the one hand by the abnormally high Verbal scores of the Indian samples and abnormally low performance scores of the same. The differences have been found to be neraly 10%. That the better Verbal scores are not due to an artifact of the adaptation has been confirmed by an experimental survey of American samples on some of the adapted subtests and by the independent authority of Torrance who made a similar observation after taking tests in Delhi some years ago. He writes :

"Incidentally the level of the children of India is comparatively higher on verbal than non-verbal tasks while the reverse is true of the US Negro sample."¹⁴

Several explanations have been advanced for the high verbal scores of Indian children but we think that the highly selected school population presents the best answer to the problem. Regarding the low Performance scores we confess to no simple or single explanation. That the performance items are associated to western play materials which are foreign to Indian children may be one reasonable explanation but certainly not a complete explanation, as it does not take into account the generally weak factor of the spatial relationship found in Indian children. The psychological homeostasis of the so-called Mirror patterns of gifted and retarded children has also been offered as an explanatory hypothesis for the disparities. We shall discuss this in the following paragraph.

Performance and Verble Tests Balanced ?

One remarkable fact stands out in the picture of score disparities discussed above, namely that the low performance scores seen to cancel out the high Verbal scores. Percentage wise comparisons on one group of data showed that where the Verbal scores were 10% higher than the American norms the Performance scores on the other hand were about 6% lower than the American norms. Could this be the result of the homeostatic functioning of the intellect hypothesized by Gallagher and Lucito in the case of the gifted?¹⁵ They write :

"Negative relationship were discovered between the gifted group and the retarded group patterns. The patterns of the mentally retarded were almost mirror image of the patterns of the gifted. The gifted appear strongest on the factor of Verbal Comprehension while they are poorest on tests relating to a Perceptual Organization factor. Exactly the opposite was true of the retarded groups."

14. E. P. Torrance, *Exceptional Children* (Journal) Washington, D. C. Sept. 1962, p. 6.

15. J. Gallagher et al, *Exceptional Children*, Washington, D. C., 1961 Vol. 27, No. 9., p. 479 seq.

CULTURAL RELATIVITY OF INTELLIGENCE QUOTIENTS.

That there is no evidence that the high verbal scores will be completely wiped out or balanced by low performance scores leaves us with the problem of a good measure of unbalanced high verbal scores pulling down our Verbal IQ's. Thus for example a 6 yrs. old American with a raw score of 9 in the Arithmetic test will get an IQ of 150 whereas an Indian will have to obtain a raw score of 12 to get the same IQ. In other words the brighter Indian mind is penalized or equated as it were with a mind less bright.

There will be no unfairness if the comparisons are made between Indian children on the same test, but since the significance of IQ levels have been given international agreement after the classical formulation of the Terman-Binet it would appear that IQ levels are not interculturally absolute. How then can we compare an American child with an IQ of 150 with an Indian child with the same statistical quotient? It would seem that the only way to make an international comparison would be to have the Indian child undertake the WISC for example with the American norms and vice versa for the American child. For those interested in such international comparisons a supplement with the American norms can be obtained.

It would also perhaps be possible to give the Indian Child a higher IQ for its better raw scores if another formula would be devised where the IQ's are not related to Percentiles or have a greater range than those presently accepted. Thus for example Instead of the 50th Percentile mean being made equivalent to an IQ of 100, it could be taken as equivalent to the 84th Percentile or an IQ of 115, and so on. But this revolutionary approach would need some international acceptance.

Computation of Scores

Having totaled the respective subtest raw scores these can be easily converted into Test Quotients (TQ's) by means of the Table accompanying this Manual.

After the conversion of raw scores into TQ's, add and find the average of each group (Verbal and Performance) separately.

Finally to obtain the Full Scale IQ, add both the Verbal and Performance test totals and divide for the average.

If only the Verbal group of TQ's are obtained they can be balanced for a Full scale IQ by adding about 6% to supply for the missing Performance scores.

6 YEARS

CONVERSION OF RAW SCORES TO PERCENTILE I.Q.'s. or T.Q.'s.

PERFORMANCE

VERBAL

Maze

Inf.	Comp.	Arith.	Simil.	Vocab.	Digit	Picture	Block	Object	Coding	Maze
R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.
1 60	1 65	1 65	1 75	1-2 55-57	1 62	1 60	1 75	1 67	1-2 55-57	1 75
2 65	2 75	2 75	2 85	3-4 60-62	2 68	2 66	2 79	2 69	3-4 59-61	2 80
3 70	3 80	3 80	3 91	5-6 65-67	3 75	3 72	3 83	3 75	5-6 63-65	3 85
4 75	4 85	4 85	4 98	7-8 70-72	4 80	4 80	4 89	4 92	7-8 67-69	4 89
5 85	5 93	5 95	5 105	9-10 75-78	5 85	5 89	5 96	5 100	9-10 71-73	5 93
6 93	6 100	6 105	6 112	11-12 80-83	6 93	6 96	6 103	6 104	11-12 75-76	6 96
7 100	7 106	7 115	7 119	13-14 85-88	7 100	7 104	7 108	7 108	13-14 77-78	7 100
8 108	8 113	8 125	8 125	15 90	8 110	8 111	8 112	8 112	15-16 79-80	8 104
9 115	9 119	9 135	9 133	16 93	9 119	9 119	9 117	9 116	17-18 81-82	9 107
10 123	10 126	10 145	10 140	17 95	10 129	10 127	10 121	10 119	19-20 83-84	10 111
11 130	11 132	11 155	11 147	18 98	11 139	11 135	11 126	11 123	21-22 85-87	11 115
12 138	12 139	12 165	12 155	19 100	12 148	12 140	12 130	12 127	23-24 89-91	12 119
13 145	13 145		12 162	20 103	13 158	13 145	13 135	13 131	25-26 93-94	13 123
14 155	14 152			21 106		14 150	14 140	14 135	27-28 96-98	14 126
15 165				22 109		15 155	15 145	15 140	29-30 100-102	15 130
				23 112			16 150	16 145	31-32 103-104	16 135
				24 115			17 155	17 150	33-34 105-107	17 140
				25 118				18 155	35-36 108-109	18 145
				26-27 120-123					37-38 111-112	19 150
				28-29 125-127					39-40 114-115	20 155
				30-31 130-132					41-42 117-120	
				32-33 134-137					43-44 122-125	
				34-35 139-141					45-46 127-130	
				36-37 143-146					47-48 132-135	
				38-39 148-150					49-50 137-140	
				40-41 153-155					51-52 142-145	
									53-54 147-150	
									55-56 152-155	

I.P.C.

M.I.S.I.C.

7 YEARS

I.P.C.

CONVERSION OF RAW SCORES TO PERCENTILE I.Q.'s. or T.Q.'s.
PERFORMANCE

VERBAL

Infor. R.S. T.Q.	Comp. R.S. T.Q.	Arith. R.S. T.Q.	Simil. R.S. T.Q.	Vocab. R.S. T.Q.	Digit R.S. T.Q.	Picture R.S. T.Q.	Block R.S. T.Q.	Object R.S. T.Q.	Coding R.S. T.Q.	Maze R.S. T.Q.
1 59	1 62	1 62	1 65	1-2 55-57	1 60	1 60	1 65	1 59	1-6 55-60	1 65
2 63	2 68	2 68	2 75	3-4 59-61	2 65	2 65	2 75	2 63	7-11 62-66	2 75
3 67	3 75	3 75	3 85	5-6 63-64	3 70	3 70	3 80	3 67	12-15 67-70	3 78
4 71	4 80	4 81	4 93	7-8 66-68	4 75	4 75	4 85	4 71	16-17 72-73	4 81
5 75	5 85	5 87	5 100	9-10 70-72	5 80	5 85	5 93	5 75	18-19 74-75	5 84
6 85	6 90	6 100	6 105	11-12 73-75	6 85	6 90	6 100	6 92	20-21 78-80	6 87
7 91	7 96	7 110	7 111	13-14 80-82	7 91	7 95	7 104	7 100	22-23 83-85	7 91
8 98	8 101	8 120	8 117	15-16 82-85	8 97	8 100	8 108	8 104	24-25 86-88	8 95
9 104	9 106	9 129	9 123	17 87	9 103	9 106	9 113	9 112	26-28 89-91	9 100
10 111	10 111	10 139	10 128	18 90	10 110	10 112	10 117	10 118	29-32 93-96	10 103
11 118	11 116	11 149	11 134	19 93	11 117	11 118	11 121	11 124	33-36 98-101	11 106
12 123	12 121	12 159	12 140	20 95	12 125	12 124	12 126	12 130	37-40 103-107	12 109
13 129	13 125		13 146	21 98	13 132	13 130	13 130	13 136	41-42 109-111	13 112
14 135	14 130		14 152	22 100	14 139	14 136	14 135	14 142	43-44 112-114	14 115
15 140	15 135		15 158	23 103	15 147	15 143	15 139	15 148	45-46 115-117	15 122
16 145	16 140		16 165	24 106	16 155	16 149	16 143	16 155	47-48 118-120	16 128
17 152	17 145			25 109	17 162	17 155	17 148		49-50 122-124	17 135
18 158	18 150			26-27 112-115			18 153		51-52 125-127	18 141
19 155	19 155			28-29 117-120			19 160		53-54 129-131	19 147
20 161				30-31 123-125					55-56 133-135	20 154
				32-33 128-131					57-58 137-139	21 160
				34-35 134-136					59-60 141-143	
				36-37 139-142					61-62 145-147	
				38-39 145-148					63-66 149-155	
				40-41 152-155						

8 YEARS

CONVERSION OF RAW SCORES TO PERCENTILE I.Q.'s. or T.Q.'s.
PERFORMANCE

VERBAL										Coding				Maze	
Infor.	Comp.	Arith.	Simil.	Vocab.	Digit	Picture	Block	Object		R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.
R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.		R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.
1 58	1 60	1 60	1 65	1-2 55-57	1 60	1 55	1 55	1 55		1-2 55-57	1 55	1 55	1 55	1 55	1 55
2 62	2 65	2 65	2 75	3-4 59-60	2 64	2 60	2 57	2 58		3-4 58-59	2 58	2 58	2 58	2 58	2 58
3 65	3 70	3 70	3 80	5-6 62-63	3 67	3 65	3 68	3 62		5-6 61-62	3 62	3 62	3 62	3 62	3 62
4 68	4 75	4 75	4 85	7-8 65-67	4 71	4 70	4 75	4 65		7-8 64-65	4 65	4 65	4 65	4 65	4 65
5 72	5 78	5 80	5 90	9-10 69-70	5 75	5 75	5 85	5 68		9-10 67-68	5 68	5 68	5 68	5 68	5 68
6 75	6 81	6 85	6 95	11-12 72-74	6 80	6 85	6 100	6 72		11-12 70-71	6 72	6 72	6 72	6 72	6 72
7 80	7 85	7 92	7 100	13-14 75-77	7 85	7 90	7 104	7 75		13-14 72-74	7 75	7 75	7 75	7 75	7 75
8 85	8 90	8 100	8 105	15-16 80-82	8 92	8 95	8 109	8 89		15-16 75-77	8 89	8 89	8 89	8 89	8 89
9 90	9 95	9 107	9 110	17-18 84-86	9 100	9 100	9 113	9 92		17-18 79-82	9 92	9 92	9 92	9 92	9 92
10 95	10 100	10 115	10 115	19-20 89-91	10 106	10 105	10 117	10 96		19-20 84-86	10 96	10 96	10 96	10 96	10 96
11 100	11 105	11 123	11 121	21 93	11 113	11 110	11 121	11 100		21-22 88-91	11 100	11 100	11 100	11 100	11 100
12 106	12 110	12 131	12 126	22 96	12 119	12 115	12 125	12 103		23-24 93-96	12 103	12 103	12 103	12 103	12 103
13 111	13 115	13 139	13 132	23 98	13 125	13 122	13 130	13 106		25 98	13 106	13 106	13 106	13 106	13 106
14 117	14 120	14 147	14 138	24 100	14 132	14 128	14 135	14 109		26 100	14 109	14 109	14 109	14 109	14 109
15 122	15 125	15 155	15 143	25 103	15 139	15 135	15 140	15 112		27 102	15 112	15 112	15 112	15 112	15 112
16 128	16 130		16 149	26 105	16 145	16 142	16 146	16 115		28 103	16 115	16 115	16 115	16 115	16 115
17 133	17 135		17 155	27 107	17 151	17 148	17 152	17 119		29 105	17 119	17 119	17 119	17 119	17 119
18 139	18 140			28 109	18 158	18 155	18 159	18 124		30 107	18 124	18 124	18 124	18 124	18 124
19 145	19 145			29-30 111-114	18 158			19 128		31 108	19 128	19 128	19 128	19 128	19 128
20 150	20 150			31-32 117-120				20 132		32-33 110-111	20 132	20 132	20 132	20 132	20 132
21 155	21 155			33-34 123-126				21 136		34-35 113-115	21 136	21 136	21 136	21 136	21 136
				35-36 129-132				22 141		36-37 118-121	22 141	22 141	22 141	22 141	22 141
				37-38 135-138				23 145		38-39 123-126	23 145	23 145	23 145	23 145	23 145
				39-40 140-143				24 150		40-41 129-132	24 150	24 150	24 150	24 150	24 150
				41-42 146-149				25 155		42-43 135-138	25 155	25 155	25 155	25 155	25 155
				43-44 152-155						44-45 141-143					
										46-47 146-149					
										48-49 152-155					

I.P.C.

M.I.S.I.C.

9 YEARS

I.P.C.

CONVERSION OF RAW SCORES TO PERCENTILE I.Q.'s. or T.Q. s.
VERBAL PERFORMANCE

Infor. R.S. T.Q.	Comp.		Arith.		Simil.		Vocab.		Digit		Picture		Block		Object		Coding		Maze	
	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.	R.S. T.Q.
1	55	1	55	1	55	1-2	55-57	1	55	1	55	1-2	55	1	55	1-3	55-57	1	55	
2	57	2	59	2	60	3-4	58-59	2	59	2	59	3	65	2	57	4-7	59-62	2	65	
3	62	3	63	3	65	5-6	61-62	3	63	3	63	4	70	3	60	8-11	63-66	3	75	
4	65	4	67	4	70	7-8	63-65	4	67	4	67	5	75	4	63	12-13	67-68	4	78	
5	68	5	71	5	75	9-10	66-67	5	71	5	71	6	85	5	65	14-17	70-73	5	82	
6	72	6	75	6	82	11-12	68-70	6	75	6	75	7	89	6	68	18-19	74-75	6	85	
7	75	7	78	7	88	13-14	71-72	7	78	7	85	8	92	7	70	20-21	77-79	7	88	
8	79	8	82	8	94	15-16	74-75	8	89	8	91	9	96	8	73	22-23	82-84	8	90	
9	83	9	85	9	100	17-18	77-78	9	100	9	97	10	100	9	75	24-25	86-89	9	93	
10	89	10	89	10	108	19-20	80-82	10	110	10	103	11	101	10	87	26-27	92-95	10	95	
11	97	11	93	11	119	21-22	83-85	11	120	11	109	12	102	11	93	28-29	97-100	11	98	
12	103	12	97	12	128	23	87	12	130	12	115	13	103	12	100	30-31	102-103	12	100	
13	109	13	102	13	137	24	90	13	140	13	120	14	105	13	104	32-33	104-106	13	104	
14	115	14	109	14	146	25	93	14	150	14	125	15	106	14	109	34-35	107-109	14	108	
15	120	15	115	15	155	26	95	15	160	15	130	16	107	15	114	36-37	110-112	15	111	
16	126	16	122	16	146	27	98	16	160	16	135	17	108	16	118	38-39	113-115	16	121	
17	132	17	128	17	152	28	100	17	152	17	140	18	109	17	123	40-41	117-118	17	127	
18	137	18	133			29	103	18	145	18	145	19	110	18	129	42-43	121-122	18	133	
19	143	19	142			30	107	19	150	19	150	20	111	19	135	44-45	126-128	19	139	
20	144	20	148			31	110	20	155	20	155	21	113	20	140	46-47	130-132	20	144	
21	155	21	155			32-33	113-116					22	115	21	145	48-49	134-36	21	150	
						34-35	120-124					23	119	22	151	50-51	138-40			
						35-37	128-131					24	123	23	156	52-53	143-45			
						38-39	135-139					25	126	24	162	54-55	147-50			
						40-41	143-146					26	130	25	168	56-57	153-55			
						42-43	150-154					27	134	26	174	58-59	157-60			
												28	137			60-61	163-65			
												29-30	142-45			62-64	167-70			

10 YEARS

I.P.C.

CONVERSION OF RAW SCORES TO PERCENTILE I.Q.'s. or T.Q.'s.
VERBAL PERFORMANCE

Infor.		Comp.		Arith.		Simil.		Vocab.		Digit		Picture		Block		Object		Coding		Maze	
R.S. T.Q.		R.S. T.Q.		R.S. T.Q.		R.S. T.Q.		R.S. T.Q.		R.S. T.Q.		R.S. T.Q.		R.S. T.Q.		R.S. T.Q.		R.S. T.Q.		R.S. T.Q.	
1	55	1	55	1	55	1	55	1-2	55-56	1	55	1	55	1-2	55-59	1	55	1-5	55-59	1	55
2	57	2	59	2	62	2	59	3-4	57-58	2	59	2	59	3-4	63-67	2	57	6-10	60-64	2	58
3	60	3	63	3	68	3	63	5-6	60-61	3	63	3	63	5-6	71-75	3	59	11-12	65-66	3	61
4	63	4	67	4	75	4	67	7-8	62-63	4	67	4	67	7-8	80-85	4	61	13-14	67-68	4	64
5	66	5	71	5	85	5	71	9-10	64-65	5	71	5	71	9-10	87-90	5	63	15-16	68-69	5	67
6	68	6	75	6	88	6	75	11-12	67-68	6	75	6	75	11-12	92-95	6	64	17-18	70-71	6	70
7	71	7	79	7	91	7	80	13-14	69-70	7	80	7	80	13-14	97-100	7	66	19-20	72-73	7	73
8	74	8	83	8	94	8	85	15-16	1-73	8	85	8	85	15	102	8	68	21-22	74-75	8	76
9	78	9	87	9	97	9	92	17-18	14-75	9	92	9	90	16	104	9	70	23-24	78-82	9	78
10	85	10	90	10	100	10	100	19-20	78-80	10	100	10	95	17	105	10	72	25-26	85-87	10	80
11	89	11	93	11	105	11	107	21-22	82-85	11	107	11	100	18	107	11	74	27-28	88-90	11	83
12	93	12	96	12	110	12	115	23-24	87-89	12	115	12	107	19	109	12	78	29-30	92-93	12	85
13	96	13	99	13	115	13	123	25-26	91-93	13	123	13	115	20	111	13	85	31-32	95-97	13	91
14	100	14	103	14	119	14	130	27-28	94-96	14	130	14	122	21	113	14	89	33-34	99-100	14	95
15	105	15	109	15	123	15	138	29-30	98-100	15	138	15	129	22	115	15	93	35-36	102-104	15	102
16	110	16	116	16	127	16	145	31-32	103-106	16	145	16	136	23	117	16	96	37-38	106-108	16	108
17	115	17	122	17	132	17	153	33-34	109-112	17	153	17	143	24	119	17	100	39-40	109-111	17	115
18	120	18	128	18	136	18	160	35-36	115-118	18	160	18	150	25	120	18	103	41-42	113-115	18	124
19	125	19	134	19	140	19	140	37-38	121-124	19	157	19	157	26-27	122-24	19	106	43-44	117-119	19	132
20	130	20	141	20	145	20	145	39-40	115-118	20	145	20	145	28-29	126-28	20	108	45-46	122-124	20	141
21	135	21	147	21	149	21	149	41-42	133-136	21	149	21	149	30-31	130-31	21	111	47-48	126-128	21	150
22	140	22	153	22	153	22	153	43-44	139-142	22	153	22	153	32-33	133-35	22	113	49-50	130-132		
23	145			23	158			47-48	151-154					34-35	137-39	23	117	51-52	135-137		
24	150							49	162					36-37	141-43	24	122	53-54	139-141		
25	155													38-39	144-46	25	126	55-56	143-145		
														40-41	148-50	26-27	131-36	57-58	148-150		
														42-43	153-55	28-29	141-45	59-60	152-155		
																30-31	150-55				

11 YEARS

I.P.C.

CONVERSION OF RAW SCORES TO PERCENTILE I.Q.'s. or T.Q.'s.
PERFORMANCE

VERBAL

Infor.		Comp.		Arith.		Simil.		Vocab.		Digit		Picture		Block		Object		Coding		Maze	
R.S.T.Q.	R.S.T.Q.	R.S.T.Q.	R.S.T.Q.	R.S.T.Q.	R.S.T.Q.	R.S.T.Q.	R.S.T.Q.	R.S.	T.Q.	R.S.T.Q.	R.S.T.Q.	R.S.T.Q.	R.S.T.Q.	R.S.	T.Q.	R.S.T.Q.	R.S.T.Q.	R.S.	T.Q.	R.S.T.Q.	R.S.T.Q.
1	55	1	55	1	55	1	55	1-2	55-57	1	55	1	55	1-2	55-60	1-2	55-56	1-2	55-56	1	55
2	58	2	59	2	60	2	59	3-4	58-58	2	59	2	58	3-4	64-68	3-4	58-61	3-5	56-58	2	58
3	60	3	62	3	65	3	63	5-7	59-61	3	62	3	62	5-6	72-75	5	63	6-9	58-61	3	62
4	62	4	65	4	70	4	67	8-10	62-64	4	65	4	65	7-8	87-89	6	65	10-13	61-64	4	65
5	65	5	69	5	75	5	71	11-13	64-66	5	69	5	68	9-10	91-93	7	67	14-18	64-68	5	69
6	67	6	72	6	78	6	75	14-15	67-68	6	72	6	72	11-12	95-96	8	69	19-24	68-73	6	72
7	70	7	75	7	82	7	80	16-18	68-70	7	75	7	75	13-14	98-100	9	71	25-27	73-75	7	75
8	72	8	78	8	85	8	85	19-20	71-72	8	85	8	81	15-16	102-103	10	73	28-29	78-81	8	77
9	75	9	82	9	89	9	90	21-22	73-73	9	93	9	89	17-18	104-106	11	75	30-31	85-87	9	79
10	82	10	85	10	92	10	95	23-24	74-75	10	102	10	100	19-20	107-108	12	81	32-33	88-90	10	81
11	87	11	89	11	96	11	100	25-26	80-85	11	111	11	106	21-22	110-111	13	86	34-35	92-93	11	83
12	90	12	92	12	100	12	105	27-28	87-89	12	120	12	113	23-24	113-114	14	90	36-37	95-97	12	85
13	93	13	96	13	104	13	111	29-30	91-92	13	129	13	119	25-26	115-17	15	93	38-39	98-100	13	90
14	97	14	100	14	107	14	117	31-32	94-96	14	138	14	126	27-28	120-22	16	96	40-41	102-103	14	95
15	100	15	104	15	111	15	123	33-34	98-100	15	147	15	132	29-30	125-28	17	100	42-43	105-106	15	100
16	103	16	108	16	115	16	128	35-36	102-104	16	155	16	139	31-32	130-32	18	103	44-45	108-110	16	107
17	106	17	112	17	120	17	134	37-38	106-108	17	160	17	145	33-34	135-37	19	106	46-47	111-13	17	115
18	109	18	117	18	125	18	140	39-40	110-108	18	182	18	152	35-36	140-43	20	109	48-49	115-17	18	121
19	112	19	122	19	130	19	145	41-42	115-117	19	198	19	158	37-38	145-47	21	112	50-51	118-20	19	128
20	115	20	126	20	135	20	152	43-44	119-122	20	210	20	188	39-40	150-52	22	115	52-53	122-24	20	135
21	119	21	131	21	140	21	158	45-46	124-27	21	220	21	198	41-42	155-57	23	119	54-55	126-27	21	141
22	123	22	136	22	145	22	165	47-48	129-31	22	230	22	208	43-44	160-62	24	123	56-57	129-31		
23	128	23	140	23	150	23	170	49-50	133-36	23	240	23	218	45	165	25	127	58-59	133-35		
24	132	24	145	24	155	24	180	51-52	138-40	24	250	24	228			26	131	60-61	137-38		
25	136	25	150					53-54	143-45							27-28	134-38	62-63	140-42		
26	141	26	155					55-56	148-50							29-30	141-45	64-65	144-46		
27	145							57-58	152-55							31-32	149-54	66-70	147-55		

12 YEARS

I.P.C.

CONVERSION OF RAW SCORES TO PERCENTILE I.Q.'s. or T.Q.'s.
VERBAL PERFORMANCE

Infor. R.S. T.Q.	Comp. R.S. T.Q.	Arith. R.S. T.Q.	Simil. R.S. T.Q.	Vocab.		Digit R.S. T.Q.	Picture		Block		Object		Coding		Maze	
				R.S.	T.Q.		R.S.	T.Q.	R.S.	T.Q.	R.S.	T.Q.	R.S.	T.Q.	R.S.	T.Q.
1-2	55-57	1 53	1 55	1-4	55-58	1 55	1 55	1-2	55-58	1-2	55-56	1-4	55-58	1 55		
3-4	59-62	2 55	2 59	5-9	58-62	2 59	2 58	3-4	62-65	3-4	57-59	5-8	59-62	2 57		
5-6	64-66	3 61	3 63	10-13	62-65	3 62	3 62	5-6	68-72	5	60	9-12	63-66	3 59		
7	69	4 64	4 66	14-17	65-68	4 66	4 65	7-8	75-77	6	62	13-16	67-70	4 61		
8	71	5 67	5 71	18-22	68-72	5 69	5 68	9-10	78-80	7	63	17-20	71-74	5 63		
9	73	6 70	6 75	23-26	72-75	6 73	6 72	11-12	82-83	8	65	21-22	75-76	6 65		
10	75	7 72	7 79	27-28	78-80	7 75	7 75	13-15	85-87	9	67	23-24	77-78	7 67		
11	80	8 75	8 83	29-30	82-85	8 80	8 85	16-18	89-91	10	68	25-26	79-80	8 69		
12	85	9 77	9 82	31-32	87-89	9 85	9 90	19-20	92-94	11	70	27-28	81-82	9 71		
13	87	10 79	10 85	33-34	90-92	10 93	10 95	21-22	95-96	12	72	29-30	83-84	10 73		
14	90	11 81	11 94	35-36	93-95	11 100	11 100	23-24	98-99	13	73	31-32	85-86	11 75		
15	92	12 83	12 104	37-38	97-98	12 107	12 107	25-26	100-101	14	75	33-34	88-89	12 78		
16	95	13 85	13 115	39-40	100-102	13 115	13 115	27-28	103-104	15	80	35-36	90-91	13 81		
17	97	14 89	14 125	41-42	103-105	14 121	14 121	29-30	105-107	16	85	37-38	93-94	14 85		
18	100	15 92	15 135	43-44	106-108	15 129	15 127	31-32	108-109	17	88	39-40	95-96	15 91		
19	104	16 96	16 145	45-46	109-110	16 136	16 133	33-34	111-12	18	91	41-42	98-99	16 97		
20	108	17 100	17 155	47-48	112-14	17 145	17 139	35-36	114-15	19	94	43-44	101-103	17 104		
21	111	18 105	18 119	49-50	115-17	18 153	18 145	37-38	117-18	20	97	45-46	105-107	18 111		
22	115	19 110	19 123	51-52	119-20	19 160	19 152	39-40	120-22	21	100	47-48	109-111	19 119		
23	119	20 115	20 127	53-54	123-25		20 158	41-42	124-26	22	104	49-50	113-15	20 127		
24	122	21 120	21 132	55-56	128-30			43-44	127-29	23	108	51-52	117-19	21 135		
25	126	22 125	22 135	57-58	132-34			45-46	131-33	24	112	53-54	121-23			
26	130	23 129	23 139	59-60	136-39			47-48	134-36	25	116	55-56	125-27			
27	133	24 134	24 143	61-62	141-43			49-50	138-40	26	121	57-58	130-33			
28-29	137-41	25 138	25 147	63-64	145-47			51-52	141-43	27-28	126-31	59-60	135-37			
30-31	145-48	26 143	26 151					53-54	145-48	29-30	135-40	61-62	140-42			
32-33	152-56	27-28 148-54	27 155					55-56	149-51			63	145			
								57-58	153-55							

13 YEARS

I.P.C.

CONVERSION OF RAW SCORES TO PERCENTILE I.Q.'s. or T.Q.'s.

VERBAL

PERFORMANCE

Infor. R.S.T.Q.	Comp. R.S.T.Q.	Arith. R.S.T.Q.	Simil. R.S.T.Q.	Vocab. R.S.T.Q.	Digit R.S.T.Q.	Picture R.S.T.Q.	Block R.S.T.Q.	Object R.S.T.Q.	Coding R.S.T.Q.	Maze R.S.T.Q.
1-2 55-57	1 55	1 55	1 55	1-3 55-57	1 55	1 55	1-2 55-58	1-3 55-57	1-2 55-57	1 55
3-4 59-61	2 58	2 58	2 59	4-5 57-58	2 58	2 58	3-4 60-63	4-7 57-60	3-4 58-59	2 57
5 62	3 60	3 61	3 63	6-9 58-61	3 62	3 62	5-6 65-68	8-9 60-61	5-6 60-62	3 59
6 64	4 63	4 64	4 67	10-12 61-63	4 65	4 65	7-8 70-73	10-11 62-63	7-8 63-64	4 61
7 66	5 65	5 67	5 71	13-16 63-66	5 68	5 68	9-10 75-77	12-14 63-65	9-10 65-66	5 63
8 68	6 68	6 70	6 75	17-19 66-68	6 72	6 72	11-12 80-83	15-17 65-67	11 68	6 65
9 69	7 70	7 73	7 78	20-23 68-71	7 75	7 75	13-15 85-87	18-20 67-69	12 69	7 67
10 72	8 73	8 77	8 82	24-28 71-75	8 80	8 80	16-20 88-92	21-23 69-71	13 70	8 69
11 74	9 75	9 81	9 85	29-30 80-85	9 85	9 85	21-23 94-96	24-25 72-73	14 71	9 71
12 75	10 77	10 85	10 88	31-32 87-89	10 92	10 90	24-26 97-99	26-28 73-75	15 73	10 73
13 79	11 80	11 92	11 91	33-34 90-92	11 100	11 95	27-28 100-101	29-30 77-79	16 74	11 75
14 83	12 83	12 100	12 94	35-36 94-96	12 107	12 100	29-31 103-105	31-32 80-82	17 75	12 80
15 87	13 85	13 107	13 97	37 98	13 115	13 106	32-33 107-108	33-36 84-87	18 87	13 84
16 90	14 88	14 115	14 100	38 99	14 121	14 112	34-37 110-13	37-40 88-91	19 89	14 88
17 93	15 91	15 122	15 104	39-43 101-105	15 128	15 117	38-39 115-16	41-42 92-93	20 92	15 92
18 96	16 94	16 129	16 107	44 107	16 135	16 123	40-41 118-20	43-44 94-95	21 95	16 96
19 99	17 97	17 136	17 111	45 109	17 142	17 129	42-43 122-24	45-47 96-98	22 97	17 100
20 102	18 100	18 144	18 115	46-48 110-12	18 150	18 134	44-45 126-28	48-49 99-100	23 100	18 104
21 106	19 105	19 150	19 119	49-50 114-15	19 156	19 139	46-47 130-31	50-51 102-104	24 104	19 115
22 109	20 111	20 157	20 124	51-52 117-19		20 145	48-49 134-35	52-53 106-108	25 107	20 122
23 114	21 116		21 128	53-54 121-23		21 151	50-51 137-39	54-55 110-112	26 111	21 130
24 117	22 122		22 132	55-56 125-27		22 156	52-53 141-43	56-58 113-117	27 115	
25 121	23 127		23 137	57-58 130-32			54-55 145-47	59-60 120-122	28 120	
26 125	24 133		24 141	59-60 134-36			56-57 150-52	61-63 124-128	29 125	
27 129	25 139		25 146	61-62 136-40			58 155	64-66 131-135	30 130	
28-29 132-35	26 145		26 150	63-64 142-44				67-68 137-140	31-32 135-140	
30-31 140-45	27 150		27 155	65-68 150-52					33-34 145-150	
32-33 150-55	28 156			69 155						

14 YEARS CONVERSION OF RAW SCORES TO PERCENTILE I.Q.'s. or T.Q.'s. PERFORMANCE

VERBAL

Infor. R.S. T.Q.	Comp. R.S.T.Q.	Arith. R.S.T.Q.	Simil. R.S.T.Q.	Vocab. R.S.T.Q.	Digit R.S.T.Q.	Picture R.S.T.Q.	Block R.S.T.Q.	Object R.S.T.Q.	Coding		Maze R.S.T.Q.
									R.S.	T.Q.	
1-2 55-57	1 55	1 55	1 55	1-3 55-57	1 55	1 55	1-3 55-59	1-2 55-59	9-10	59-60	1 55
3-4 59-60	2 57	2 58	2 59	4-6 57-59	2 58	2 58	4-7 62-68	3-4 58-59	11-12	60-61	2 57
5 62	3 59	3 61	3 62	7-9 59-61	3 61	3 62	8-10 71-75	5-6 60-61	13-14	61-62	3 59
6 64	4 62	4 63	4 65	10-12 61-63	4 63	4 65	11-12 76-77	7-8 62-63	15-16	62-63	4 60
7 66	5 64	5 66	5 68	13-15 63-65	5 66	5 68	13-16 79-82	9 65	17-18	63-64	5 62
8 68	6 66	6 69	6 72	16-18 65-67	6 69	6 72	17-19 84-86	10 66	19-20	64-65	6 64
9 70	7 68	7 72	7 75	19-21 67-59	7 72	7 75	20-21 88-89	11 67	21-22	65-66	7 65
10 71	8 71	8 75	8 78	22-24 69-71	8 75	8 78	22-24 91-93	12 68	23-24	66-67	8 67
11 73	9 73	9 79	9 82	25-26 71-72	9 85	9 82	25-26 95-96	13 69	25-26	67-78	9 68
12 75	10 75	10 83	10 85	27-28 72-73	10 92	10 85	27-28 98-99	14 71	27-29	68-70	10 70
13 77	11 77	11 89	11 89	29-31 73-75	11 100	11 90	29-32 101-104	15 72	30-31	70-71	11 72
14 79	12 79	12 97	12 93	32-33 78-82	12 108	12 95	33-38 106-111	16 73	32-33	71-72	12 73
15 81	13 81	13 104	13 96	34-35 85-87	13 115	13 100	39-41 113-115	17 75	34-35	72-73	13 75
16 83	14 83	14 112	14 100	36-37 89-91	14 122	14 105	42-43 117-118	18 78	36-37	73-75	14 80
17 85	15 85	15 120	15 104	38-39 93-94	15 130	15 111	44-45 120-121	19 82	38-44	77-89	15 85
18 89	16 89	16 129	16 107	40-41 95-96	16 138	16 117	46-47 123-124	20 85	45-46	90-92	16 92
19 92	17 94	17 136	17 111	42-43 100-101	17 145	17 122	48-50 126-128	21 88	47-49	93-97	17 100
20 96	18 98	18 144	18 115	44-45 103-104	18 153	18 128	51-52 130-131	22 92	50-51	98-100	18 107
21 100	19 103	19 152	19 119	45-47 106-107	19 161	19 134	53-54 133-134	23 96	52-53	101-103	19 115
22 105	20 110	20 160	20 123	48-49 109-110		20 139	55-56 136-137	24 100	54-55	104-105	20 122
23 111	21 116		21 128	50-51 112-113		21 145	57-58 139-140	25 105	56-57	107-108	21 130
24 116	22 122		22 132	52-54 115-119		22 151	59-60 142-143	26 110	58-59	110-111	22 137
25 122	23 129		23 136	55-60 120-132		23 157	61-62 145-150	27 115	60-62	113-115	23 145
26 127	24 135		24 140	61-63 132-136				28 121	63-64	117-119	24 152
27 132	25 141		25 145	64-69 137-147				29 127	65-68	122-128	25 160
28-29 138-43	26 148		26 150	70-71 150-152				30 134	69-71	131-135	
30-31 149-45	27-28 154-60		27-28 155-160	72 155				31-32 140-145	72-73	138-140	
								33-34 151-157			

15 YEARS

CONVERSION OF RAW SCORES TO PERCENTILE I.Q.'s. or T.Q.'s.

VERBAL

Infor.		Comp.		Arith.		Simil.		Vocab.		Digit		Picture		Block		Object		Coding		Maze	
R.S.	T.Q.	R.S.	T.Q.	R.S.	T.Q.	R.S.	T.Q.	R.S.	T.Q.	R.S.	T.Q.	R.S.	T.Q.	R.S.	T.Q.	R.S.	T.Q.	R.S.	T.Q.	R.S.	T.Q.
1-2	55-57	1	55-57	1	55	1	55	1-3	55-57	1	55	1	55	1-2	55-57	1-2	55-56	15-16	62-63	1	55
3-4	59-60	3-4	59-60	2	58	2	58	4-6	57-59	2	58	2	58	3-4	58-60	3-4	58-59	17-18	63-64	2	57
5-6	62-63	5-6	62-63	3	61	3	61	7-10	59-62	3	61	3	61	5-6	61-63	5-6	60-61	19-20	64-65	3	59
7-8	65-67	7	65	4	63	4	63	11-13	62-64	4	63	4	64	7-8	64-66	7-8	63-64	21-22	65-66	4	60
9-10	68-70	8	67	5	66	5	66	14-17	64-67	5	66	5	66	9-10	67-69	9-10	65-66	23-24	66-67	5	62
11	72	9	68	6	65	6	65	18-20	67-69	6	69	6	69	11-12	70-72	11-12	67-69	25-26	67-68	6	64
12	73	10	70	7	72	7	72	21-24	69-72	7	72	7	72	13-14	73-75	13	70	27-29	68-70	7	65
13	75	11	72	8	75	8	75	25-28	72-75	8	75	8	75	15-17	77-79	14	71	30-31	70-71	8	67
14	78	12	73	9	80	9	78	29-30	77-78	9	83	9	85	18-19	81-82	15	72	32-33	71-72	9	69
15	82	13	75	10	85	10	82	31-33	80-82	10	90	10	89	20-22	84-86	16	74	34-35	72-73	10	70
16	85	14	80	11	90	11	85	34-36	84-88	11	98	11	92	23-25	88-90	17	75	36-38	73-75	11	72
17	88	15	85	12	95	12	89	37-40	91-97	12	104	12	96	26-27	92-93	18	86	39-43	77-85	12	74
18	90	16	89	13	100	13	94	41-44	100-103	13	112	13	100	28-29	95-96	19	91	44-45	86-88	13	75
19	92	17	93	14	106	14	98	45-48	105-108	14	118	14	105	30-31	98-99	20	95	46-47	89-90	14	80
20	95	18	96	15	114	15	102	49-52	109-112	15	125	15	110	32-35	101-104	21	100	48-49	92-93	15	85
21	97	19	100	16	121	16	106	53-56	113-116	16	132	16	115	36	106	22	103	50-51	95-96	16	92
22	100	20	105	17	129	17	111	57-58	118-119	17	139	17	120	37	107	23	106	52-53	98-99	17	100
23	103	21	111	18	136	18	115	59-60	120-121	18	146	18	125	38	108	24	110	54-58	101-105	18	107
24	107	22	116	19	143	19	120			19	153	19	130	39-42	110-113	25	113	59-62	107-110	19	115
25	110	23	122	20	151	20	125			20	160	20	135	43-44	115-116	26	116	63-65	112-114	20	122
26	113	24	127	21	158	21	130			21	160	21	143	45-47	118-120	27	120	66-67	116-117	21	130
27	117	25	133			22	135			22	160	22	146	48-59	122-123	28	123	68-69	119-120	22	138
28-29	120-24	26	138			23	140			23	160	23	151	50-51	125-126	29-30	126-130	70-72	122-124	23	146
30-31	128-31	27	144			24	145			24	160	24	157	52-54	128-130	31-32	133-136	73-74	126-127	24	154
32-33	135-38	28	149			25	150			25	160			55-58	132-138	33-34	140-143	75-76	129-130		
34-35	142-46	29	155			26	155			26	160			59-62	140-146	35-36	146-150	77-78	132-133		
36-37	150-54					27	160			27	160			63-64	149-151	37-38	153-156	79-80	135-137		
														65-66	152-155			81-90	138-156		
														67-68	158-160						

INTELLIGENCE SCALE FOR INDIAN CHILDREN (ISIC)

Dr. A. J. Malin M.A., M.Ed., Ph.D.

ADAPTATION OF WISC

DIRECTOR FOR ADMINISTRATION OF THE ISIC

1. INFORMATION TEST

Directions : Read each question as stated and in the order given, If responses are not clear it is permissible to say : Explain more : tell me more about it : or some such neutral non-leading questions. Question 1-5 are used for those below 8 years or suspected mental defectives. Acceptable answers are indicated in Parentheses.

Scoring: Discontinue after five consecutive failures and credit with items 1-5 if subject passes 6-7 and 8. Each item is scored 1 or 0.

TEST QUESTIONS

1. How many ears have you ?
 2. What do you call this finger (show thumb)?
 3. How many legs has a dog?
 4. What animal gives us milk?
 5. How do we make boil ? (Apply heat, etc.)
 6. Where will you go to buy sugar (ribbons-girls)? (market, kirana, local shop's name etc.)
-
7. How many paise in a rupee?
 8. How many days in a week?
 9. Where does the sun set?
 10. What are the seasons of the year (in India)? (at least cold, hot, rainy)
 11. Which is the top or upper colour of the Flag of our country? (red, orange etc.)
 12. When do we celebrate (keep) Independence Day? (if too close, substitute:
What do we celebrate on Janury 26 ?
 13. What is the colour of the pearl? (white)
 14. What is the work of the stomach? (mixes, grinds, digests).
 15. Why do we honour Gandhiji as the Father of India? (Independence, labours.)

16. Why does oil float on water? (lighter)
17. What was the capital of old Mogul empire. (Agra, Delhi)
18. What is Sanskrit? (imply ancient or sacred Indian language)
19. How tall is the average (ordinary) Indian man? (5ft. 4 inches-3" leeway)
20. In what part of the world do we find India? (Asia; Northern or Eastern Hemisphere).
21. Who composed the national anthem (Jana gana mana)?
22. Where is Singapore ? (Malaya : S.E. Asia; near Borneo, Australia, etc.)
23. How many thousands are there in a lakh?
24. Who was the first European or foreign invader to come into India?
(Alexander, Vasco da Gama)
25. What is the distance between Bombay and Calcutta? (1000-300 miles)
26. What is a Barometer ? (measures airpressure, heights; predicts weather)
27. What (or, why) do we celebrate on Republic Day? (Constitution, abolition of monarchy, Notion of Independence not enough)
28. How do we get kerosene? (imply process from crude oil)
29. Who was Chenghis Khan? Chinese or Mongol-not Mogul-Ruler)
30. What is the meaning of V.P.P.? (Value payable parcel, or a correct description of it.)

II GENERAL COMPREHENSION TEST

Directions : Repeat questions if needed and encourage with remarks such as "Yes, Go ahead" and in case additional credited reasons are expected as in 6-14 "Yes, and also because"

Scoring : Discontinue after 3-consecutive failures. Scores are graded from 2, 1 to 0. See criteria scoring samples. (below.)

TEST QUESTIONS

1. What should you do if you cut your finger? For example. while sharpening a pencil.)
2. What would you do if your Mother sends you to buy sugar and the shopkeeper says he does not have anymore?

-
3. What would you do if a boy (girl) smaller than you starts to fight with you ?
 4. What should you do if through your own fault the ball (doll) of your friend is lost and you cannot find it?
 5. What should you do if you found a railway line broken and at the same time noticed a train about to come?
-

6. Instead of spending all your money, why would it be better to save some, for example in a bank?
7. What good is it to send criminals to jails to prisons?
8. Why does the small shopkeeper prefer cash to giving credit?
9. Why should we respect and obey those who teach us?
10. Why should we be kind to animals?
11. Why should Government set examinations when selecting job applicants?
12. Why is bad to break a promise?
13. Why is it wrong to steal?
14. Why is a Parliament needed in our system of Government?

CRITERIA AND SCORING SAMPLES

Items : 1-5 are scored 2 point if Subject assumes personal responsibility for what is done. Allow 1 point only if responsibility is shirked but S. knows what is to be done. Items 6 - 14 allow 2 points for 2 good reasons and 1 point if only one can be given. Do not allow religious reasons.

1. Cut : 2 = apply medicine ; cleanse; etc. 1 = Sec Doctor. 0 = Cry.
2. Sugar : 2 = Go to another ; Go to another ; 1 = Go home and tell. (Q)
3. Fight : 2 = Dont hit back ; report 0 = hit back
4. Ball : 2 = replace. 1 = Say Sorry (Q)
5. Train : 2 = Some kind of warning 1 = Tell Station Masters 0 = Tell police.
6. Bank : Safety ; interest ; future needs; encourage thrift

7. Jails : Stop evil ; punish ; community safety ; teach lesson.
8. Cash : Small customer forgets; cheated.
9. Teachers ; Elders; place of parents; help us ; wiser.
10. Animals : Dumb and helpless ; useful ; sensitive like ourselves.
11. Jobs : Select fit ones ; give equal chance to all.
12. Promise : Lose repute; spoil character; harm others.
13. Steal ; Unjust possession and profit; harmful deprivation.
14. Parliament : Popular representation ; brings Government closer.

III ARITHMETIC TEST

Directions : Problems 1, 2, 3 are for subjects below 8 yrs or suspected mental defectives. Problems 4-12 and 16 are read to the subject and 13, 14, 15 are presented on cards to be read by the subject. Timing starts after stating the problem the first time. Repetitions are at the expense of the subject's timing. Discontinue after 3 consecutive failures. No paper work allowed. Names and problem items can be adapted to local conditions and prices. Figures must not be changed. Second trials allowed within time limit.

Scoring : 1 or 0. Give credit for first three if subject works out 4 and 5.

PROBLEMS - TIMINGS - ANSWERS

- | | |
|---|-----|
| 1. Place 9 blocks (or coins) in a row saying : "Count these with your finger" | 45" |
| 2. Rearrange the row saying : "Give 4 blocks to me" | 45" |
| 3. Rearrange as above saying : " Take 7 and give me the rest" | 45" |
| <hr/> | |
| 4. If I break this (pencil) in half, how many pieces will there be ? 2 | 30" |
| 5. Prem has 4 rupees. Mummy gives him 2 more. How many has he? 6 | 30" |
| 6. Rita has 8 bananas and buys 6 more. How many has she? 14 | 30" |
| 7. Teacher has 12 books and sells 5. How many has she left? 7 | 30" |
| 8. If one pencil costs 7 p. what will 3 pencils cost ? 21p | 30" |

9. A Milkman has 25 bottles of milk, he sold all. How many bottles had he left? 0 30
10. A workman after finishing his job was given Rs. 36 for his pay. He had agreed for Rs. 4 each day. Tell me how many days he worked. 9 30
11. Let us say you want to buy some marbles (ribbons). They cost 30 (make this distinct from 13) paise a dozen. Now you have a rupee and want to buy 3 dozen. How much change will you receive? (10p.) 70p 60
12. Four boys (girls) have gathered 72 marbles (flowers). If they divide them equally, how much will each get? (18) 18 30

(Present the following 3 problems on separate cards to be read)

13. If 3 pencils cost 5 paise, what will 24 pencils cost? (40) 60
14. If bus fare is 20 paise for the first quarter mile and only 5 paise for each other quarter mile; what will be the fare for 2 miles? (55) 120
15. Prem and Raj start a marble game with 27 marbles each They agree that at the end of each game, the loser must give up one third of what he has left. 120
- Now, Prem wins three games. How much has Raj left for the 4th game? (8) 120
16. 36 is two-thirds of what whole number? (54) 30

IV ANALOGIES AND SIMILARITIES

Direction Analogies :- For subjects under 8 years. Before reading each item say : "Now listen carefully as I am going to read you a sentence which you must finish." Score 2 for each correct analogy.

ANALOGIES TEST

1. Lemon are sour, but sugar is
2. You walk with your legs and throw with your (hand , arm)
3. Boys grow up to be men and girls grow up to be
4. A knife and a piece of glass can both

SIMILARITIES TEST

Directions :- Say : "In what way are a mango and a banana alike?" If the Subject fails or denies a similarity then coach him on all score grades and try the next Cat and Mouse. If he fails again explain for the last time and discontinue after 3 consecutive failures, or return to Analogies.

Scoring :- Grade from 2 to 1 and 0. See Below for Grades. Give credit for 4 analogies if subject earns at least 3 points on Similarities.

TEST ITEMS

5. Mango-Banana (or other familiar fruits).
6. Cat - Mouse
7. Organ - Flute (Similar instruments such as a Harmonium or mouth organ may be substituted).
8. Milk - Medicine.
9. Ruler (e.g. foot) - Scale (weights).
10. Scissors - brass pot (vasé).
11. Paper - Coal (charcoal).
12. Salt-Water.
13. Mountain-Lake
14. Liberty-Justice.
15. First - Last
16. Numbers 49 and 121.

GRADES AND SCORING SAMPLES

5. Mango : 2 = fruits 1 = both round; have skin; seeds; food. 0= Sweet, from tree.
6. Cat : 2 animals creatures; 1 = four legs; eyes; both eat. 0 = chase
7. Organ : 2 = both wind or musical instruments 1 = play them both ; have keys; tubes. Both give tunes. 0 = both give noise.

8. Milk : 2 = both are good for health; good to take; 1 = consumables (drink)
9. Ruler : 2 = both measure ; 1 = both have numbers 0 = made of same thing.
10. Scissors : 2 = both made of metal utensil. 1 = made of iron, steel.
11. Paper : 2 = carbons ; originate from trees. 1 = both burn;
12. Salt : 2 chemical compound; necessary for life. 1 = for cooking.
13. Mountain : 2 = geographical or natural features of landscape; common origin.
1 = both are scenery, landscape.
14. Liberty : 2 = social ideal, or rights, 1 = relate to Government : needed for the country. have to do with law 0 = mean peace.
15. First : 2 = extremes of position ; position in a series or rank. 1 = both at the ends.
0 = both opposites ; both numbers
16. 49 : 2 = perfect squares or odd number square roots. 1 = both odd numbers;
cannot be divided by 2. 0 = Both are numbers.

V. VOCABULARY TEST

Directions :- Say : "I want to see how many words you know. Tell me now for example what CYCLE means. What is a cycle?" If response is vague or has only the score value of 1 then question further, (Q) for example, "What else does it mean? Explain a little more"

Scoring :- Discontinue after 5 consecutive failures. Each word is scored 2, 1 or 0 except words, 1 through 6 are scored only 2 or 0. Criteria and examples of marginal responses are found below. Subjects may be started with 10 and credited with 2 points for the previous unless subject fails to give 2 point definitions from 10 to 14, in which case proceed backwards until 5 consecutive 2 point successes are met, then return and continue from the first failed 2 pt.

TEST WORDS

- | | | | |
|-------------------|--------------|----------------|--------------------|
| 1. Cycle | 11. Hammer | 21. Martyr | 31. Bail |
| 2. Knife | 12. Mischief | 22. Editor | 32. Apprentice |
| 3. Shoe (Chappal) | 13. Gold | 23. Ambassador | 33. Affliction |
| 4. Umbrella | 14. Meet | 24. Auction | 34. Boycott |
| 5. Pillow | --- | 25. Stanza | 35. Aseptic |
| 6. Letter | 15. Clever | 26. Proverb | 36. Nitroglycerine |
| --- | 16. Diamond | 27. Anonymous | 37. Spangle |

7. Nail	17. Brave	28. Shameful	38. Hara-kiri
8. Donkey	18. Shilling	29. Buoy	39. Mantis
9. Leather	19. Gambling	30. Atheist	44. Sedition
10. Sword	20. Microscope		

VOCABULARY GENERAL CRITERIA

A. 2 point responses can be, a good synonym or translation; a major use e.g. Donkey-carries stones, laundry : one or more Primary features : e.g. long ears; small horse : a genetic classification, e.g. animal or finally less definitive but correct cumulative description e.g. four legs, very short, brays, etc.

B. 1 point responses can be vague synonyms e.g. Nail-tool; a minor use not elaborated e.g. sword-something to cut with ; a correct but non-distinguishing attribute e.g. donkey got tail or giving an example of the use of the word e.g. Nail-pounding the nail into the wall; hanging a calendar on it.

SAMPLE RESPONSES

1. Cycle : 2 = What you ride on. It has wheels, etc.
2. Knife : 2 = To cut bread. Is sharp; has a point
3. Shoe : 2 = Wear it. Made of leather. A boot.
4. Umbrella : 2 = For rain, sun. Put over the head. Is round with a handle.
5. Pillow : 2 = To rest the head; for sleeping. Has cotton inside.
6. Letter : 2 = Post card. Write to friend, We send it. Postman brings it.
0 = Writing. Paper with writing
7. Nail : 2 = pound into wood; has a point, On our finger.
1 = It is iron, sharp, Used to hang calendars. Pound into walls.
8. Donkey : 2 = Small horse; carries clothes, stupid animal, animal; brays.
1 = like a horse; got a tail;
9. Leather : 2 = skin; makes shoes; from animals.

10. Sword : 2 = Fight with it ; it kills; has a point; like a knife.
1 = to cut with ; is sharp, 0 = cuts bread.
11. Hammer : 2 = to pound with ; tool ; iron with handle.
1 = to hammer ^{nails} ~~nails~~ ; carpenter uses it ; breaks coal.
12. Mischief : 2 = naughty : fooling; trouble someone; describe such an act.
1 = very bad;
13. Gold : 2 = precious metal : is very valuable ; make rings, ornaments of it.
1 = It shines; a gold necklace.
14. Meet. : 2 = Join; together; to see, visit, greet someone
1 = to meet a friend; to play with one.
0 = meet somebody; a meeting
15. Clever : 2 = Smart ; cannot be tricked or fooled. very tricky.
2 = Bright, intelligent, good in school, good brain.
16. Diamond : 2 = sparking, shiny stone, very valuable, piece of jewelry; hardest substance, carbon.
1 = glass ; diamond ring; play cards with, square shape, shines.
0 = stone, metal.
17. Brave : 2 = Can fight everyone and in wars;
1 = brave in fighting ; strong and clever ; tiger is brave.
18. Shilling : 2 = English coin ~~or~~ money; part of a pound.
1 = Foreign, American, olden money.
0 = silver coin, a rupee, money.
19. Gambling: 2 = play with money ; win on luck; to bet.
1 = waste money; gamble money ; 0-to play cards; it is bad.
20. Microscope : 2 = to see germs and insects; makes them big (many items).
1 = something to see.

21. Martyr : 2 = dies for faith or country. Gandhiji. 0 = Saint.
22. Editor : 2 = Selects for publication or writes editorials,
1 = Publishes, prints, writes for newspaper, head of newspaper.
0 = Writes (Q).
23. Ambassador : 2 = Sent by King to make peace; carries messages to another country; represents. 1 = Does work in another country :
24. Auction : 2 = Sale by bidding;
1 = Public sale; selling old things; American auction. 0 = Selling.
25. Stanza : 2 = Part of poem ; verse; part of a song ; paragraph in poetry.
1 = A poem; some lines; a paragraph.
26. Proverb : 2 = A saying ; sentence with a moral ;
1 = Sentence from olden time ; from the Bible;
27. Anonymous : 2 = Without name of author. 1 = no name; unknown.
28. Shameful : 2 = A low feeling; want to hide; made shy; can't be proud of.
1 = a bad thing ; 0 = thing full of shame.
29. Buoy : 2 = Float to warn ships or mark places; to save from drowning.
1 = Bell-buoy; something floating; 0 = Life-Buoy soap.
30. Atheist : 2 = Unbeliever; without religion. 1 = Does not pray or believe in Christ.
31. Bail : 2 = Jail release money: used in cricket; take out water.
1 = money.
32. Apprentice : 2 = Learns trade; 1 = Assistant.
33. Affliction : 2 = Sorrow, hurt. 1 = afflicted with disease; infection.
34. Boycott : 2 = to stop buying from or going to a place out of protest.
1 = to put out (British).
35. Aseptic : 2 = without germs 1 = medicine to cleanse : not septic
36. Nitroglycerine : 2 = Explosive. 1 = dangerous chemical.
37. Spangle : 2 = sort of sequin, shiny thing on clothes. 1 = shiny ornament.

38. Hara-Kiri : 2 = Jap method of suicide. 1 = to kill in Japan or China.
 39. Mantis : 2 = Insect - like grasshopper. 1 - An insect.
 40. Sedition : 2 = Excite rebellion to Government. 1 = Break laws.

VI DIGIT SPAN TEST (Alternate)

DIGIT SPAN FORWARDS

Directions : " I am going to say some numbers. Listen carefully and when I am finished repeat them after me" (one per second). If subject repeats trail I of a series then go on to next higher number in same Trial. If he fails then give a second chance from Trail II. Discontinue if fails on both Trails of a given series.

Scoring : His score is the highest number of digits repeated without error. Thus if he only repeated five digits his score is 5. Total score combines forward and backward.

Series	Trial I	Trial II
3	3-8-6	6-1-2
4	3-4-1-7	6-1-5-8
5	8-4-2-3-9	5-2-1-8-9
6	3-8-9-1-7-4	7-9-6-4-8-3
7	5-1-7-4-2-3-8	9-8-5-2-1-6-3
8	1-6-4-5-9-7-6-3	2-9-7-6-3-1-5-4
9	5-3-8-7-1-2-4-6-9	4-2-6-9-1-7-8-3-5

DIGIT SPAN BACKWARDS

Directions : "Now I am going to say some more number but this time when I stop you say them backwards." Give an example and let the subject try. Scoring and directions as in the above.

Series	Trial I	Trial II
2	2-5	6-3
3	5-7-4	2-5-9

4	7-2-9-6	8-4-9-3
5	4-1-3-5-7	9-7-8-5-2
6	1-6-5-2-9-8	3-6-7-1-9-4
7	8-5-9-2-3-4-2	4-5-7-9-2-8-1
8	6-9-1-6-3-2-5-8	3-1-7-9-5-4-8-2

VII. PICTURE COMPLETION

Directions :- Before presenting the first card say : "I am going to show you some pictures in which there is a part missing. Look carefully at the card and tell me what important part is missing." The Subject can be helped in the first two cards but without getting a score. Pointing to the missing part can also be credited. If S. point to, e.g. coat pocket, say "Yes, but something else is missing."

Timing and Scoring :- Discountinue after 4 failures. 15 seconds are allowed for each picture, and one point for each correct response except with the last five pictures when an extra bonus score is credited if at least 3 of the last five are correct. Total maximum score 20.

TEST PICTURES AND CORRECT RESPONSES

- | | |
|-----------------------------|---------------------------------------|
| 1. Comb tooth (teeth) | 11. Rooster spur |
| 2. Fox ear | 12. Screw slot (crack) |
| 3. Girl mouth | 13. Fish extra fin (dorsal) |
| 4. Table leg | 14. Fly antennae |
| 5. Cat whiskers | 15. Profile eyebrow |
| 6. Hand fingernail | 16. Cow hoof cleft |
| 7. Scissors screw | 17. Umbrella spokes |
| 8. Coat buttonholes | 18. Thermometer mercury in bulb |
| 9. Door hinge | 19. House tree shadow. |
| 10. Card centre spade | 20. Man |

VIII. BLOCK DESIGN

Directions : For Subjects under 8 years.

Demonstrate Design A, then take 4 more blocks and tell subject to imitate (Not from card!) If fails, then correct (i.e. demonstrate) a second time with blocks of S. Score 2 pts. if first trial correct and 1 pt., if only second trial was passed. Arrange Design B behind a screen and proceed as in A. With design C, use Card and demonstrate without screen explaining how it agrees with the card. Then brush up pile for S to begin. Score as with A. Timings for A-C : 45."

Directions : For Subjects under 8 years and above

Take blocks say : "You see these blocks have different colours and they can be put together in a picture like this." Give four blocks to the Subject brushed up so that only one red colour faces upwards. Then say : "Now I am going to show you a picture made of red blocks and I want you to try to copy it as quickly as you can" Then show Card C and start timing. Second trial may be given on C also, but without credit. On designs 1-7 discontinue after 2 consecutive failures.

Scoring :- Advance credit of 4 points is given for designs A., and B and 2 pts. more for passing C. Design C. Designs 1-7 give 4 points plus bouns or per table below. No points are given incomplete design 5-7 are made with 9 blocks and 2 brushed up red colours.

Design 1	75".....	1-10" (7 points).....	11-15" (6 points)	16-20" (5 points)
„ 2	 „	1-10" („).....	11-15" („)	16-20" („)
„ 3	 „	1-15" („).....	16-20" („)	21-25" („)
„ 4	 „	1-10" („).....	11-15" („)	16-20" („)
„ 5	150".....	1-35" („).....	36-45" („)	46-65" („)
„ 6	 „	1-55" („).....	56-69" („)	70-80" („)
„ 7	 „	1-55" („).....	56-60" („)	61-90" („)

IX OBJECT ASSEMBLY

Directions : Bouns allowed for speed. See table below :

MANIKIN	120".....	1-10" (7pts.)	11-15" (6pts.)	16-20" (5pts.)	21-120" (4pts.)
Horse	180".....	1-15" (9pts.)	16-20" (8pts.)	21-30" (7pts.)	31-180" (6pts.)
Face	 180".....	1-35" („)	36-45" („)	46-70" („)	71-180" (6pts.)
Auto	 180".....	1-25" („)	26-30" („)	31-45" („)	46-180" („)

MANIKIN : Directions :- Arrange pieces behind a screen according to given diagram. Then say : "These pieces will make a boy. Go ahead and put them together."

Scoring Table :- Perfect performance - 4 pts. or bouns.

Imperfect performances : NO TIME bouns ! 2. pts. if legs omitted or put as arm.

3 pts. if legs interchanged or inverted. 1. 1 pt. if only trunk is correct.

HORSE : Directions same as for Manikin.

Scoring Table : Perfect - 6pts. or bonus.

Imperfect 5 pts. if only stomach midpiece inverted

4 pts. if midpiece omitted or legs interchanged. 3 pts. if midpiece inverted and legs :
2 pts. if : omitted and : or 1 & 4. 1 pt. for each two peices joined properly.

FACE : Directions same as above EXCEPT no name of the object is revealed.

Scoring Table : Perfect - 6 pts. or bouns.

Imperfect : General 1/2 pt. for each proper joint seperate or joined to the whole.

5 pts. if eye inverted or hair pieces omitted. 4 pts. if nose omitted or mouth and chin
omitted. 3 pts. if large half omitted (2 pieces) 1 pt. if only large half and hair (3pieces).

AUTO : Directions same as for Face.

Scoring Table : Perfect - 6 pts. or bouns.

Imperfect : General 1 pt. for each proper joint as above.

5 pts. Door inverted or reversed. 4 pts. Omitting pieces 4 & 5 or omitting 7 3 pts.
Omitting 7 and inverting or reversing 4 (door) also omitting 4, 5, 7. also interchanging 4 & 5
with 6.

X CODING

Directions : for Subject under 8 years and suspected mental defectors,

Use Design A. Point to the row of stars, squares, etc, and say; "Look at these and see
how each one has some mark inside. Now look at these samples and I want you to put in
each figure the same mark that you find here in the top row. Here is a star so put this mark
inside." After marking the first two or there samples let the subject fill in the remaining and
then start timing when the samples have been finished.

Timing & Scoring :- 120 time limit and the score is the number of designs excluding the samples completed in that time. If finished before, then the following bonus table is used :

101 — 110 — 46 pts.	71—80 — 49 pts.
91 — 110 — 47 pts.	70 or less — 50 pts.

Directions :- for Subjects 8 years and above.

Use Design B Point to the key and say :- "Look and these boxes or square. See how each has a number in the upper half and a mark in the lower half. Each number has its own special mark. Now look at the samples. Here is a 2. so put in this mark (here write the symbol) and try the others until I tell you to stop." Start timing when the samples have been completed. Watch out that the subject does not skip.

Timing & Scoring :- 120 seconds and 1 point for each correct square excluding the samples.

Note :- For left hand subjects an extra folded code so that they do not obscure their vision.

XI MAZES

Directions : Show the sample and with a pencil demonstrate saying : "Imagine someone trying to get out of here. He cannot cross the lines and you cannot lift the pencil once you start" Subjects above 8 years can start with C and be credited for A and B if not more than one error on C. Subjects below 8 years start with A. if fails on A or B then show correct procedure. Discontinue if 2 consecutive failures (0 scores). Timing printed beside each maze .

Scoring : A, B, C. - No errors - 2 pts. Not more than 2 errors - 1 pt.

1-5. - No errors - 3 pts. First error subtracts one point and the second error subtracts the second point but the third point is not subtracted until the maximum allowed errors have been passed, viz.,

Nos. 1—2 allowed 3 errors

3	„	5	„
4	„	6	„
5	„	8	„

Definition of Errors :

1. Entering a major blind alley (wandering around in such a blind alley including lesser blind alleys does not increase the error nor or reentering it after going out).
2. Crossing any line. A white space should be visible between the printed line and the pencil line on the wrong side.
3. Lifting the pencil Do not penalize twice if the lifting was done for example to return to the path after crossing a line. In which case the error of crossing the line is alone counted.

Record Form

Malin's Intelligence Scale For Indian Children (MISIC)

NAME.....SEX.....

ADDRESS.....

FATHER'S NAME.....MOTHER'S NAME.....

SCHOOL.....CLASS.....

	YEAR	MONTH	DAY		RAW SCORE	I/Q
Date tested				Verbal Scale		
Date of Birth				Performance Scale		
Age				Total		

VERBAL TABLE	Raw Score	I.Q.	PERFORMANCE TEST	Raw Score	I/Q
Information			Picture Completion		
Comprehension			Block Assembly		
Arithmetic			Object Assembly		
Vocabulary			Mazes		
(Digit Span)					
Sum of Verbal Tests			Sum of Performance Test		

1. INFORMATION TEST

Response	Score	Response	Score
1.		16.	
2.		17.	
3.		18.	
4.		19.	
5.		20.	
6.		21.	
7.		22.	
8.		23.	
9.		24.	
10.		25.	
11.		26.	
12.		27.	
13.		28.	
14.		29.	
15.		30.	

IV. ANALOGIES # SIMILARITIES TEST:

	Response	Score
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		

	Response	Score
9.		
10.		
11.		
12.		
13.		
14.		
15.		
16.		

11. GENERAL COMPREHENSION TEST:

	Response	Score
1.		
2.		
3.		
4.		
5.		
6. a)		
b)		
7. a)		
b)		
8. a)		
b)		

Total

	Response	Score
9. a)		
b)		
10. a)		
b)		
11. a)		
b)		
12. a)		
b)		
13. a)		
b)		
14. a)		
b)		

1. ARITHMETIC TEST:

Problem	Response
1. 45"	
2. 45"	
3. 45"	
4. 30"	
5. 30"	
6. 30"	
7. 30"	
8. 30"	
9. 30"	
10. 30"	
11. 60"	
12. 30"	
13. 60"	
14. 120"	
15. 120"	
16. 30"	

Total

Score	Design	Time Pass-Fail	Score
A. 45"			
B. 45"			
C. 45"			
1. 75"			
2. 75"			
3. 75"			
4. 75"			
5. 150"			
6. 150"			
7. 150"			

Total

VI. DIGIT SPAN

FORWARDS		BACKWARD	
Response	Score	Response	Score
3. a)		2. a)	
b)		b)	
4. a)		3. a)	
b)		b)	
5. a)		4. a)	
b)		b)	
6. a)		5. a)	
b)		b)	
7. a)		6. a)	
b)		b)	

V. VOCABULARY TEST

Response	Score	Response	Score
1.		21.	
2.		22.	
3.		23.	
4.		24.	
5.		25.	
6.		26.	
7.		27.	
8.		28.	
9.		29.	
10.		30.	
11.		31.	
12.		32.	
13.		33.	
14.		34.	
15.		35.	
16.		36.	
17.		37.	
18.		38.	
19.		39.	
20.		40.	

VII. PICTURE COMPLETION

	Response	Score
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		
11.		
12.		
13.		
14.		
15.		
16.		
17.		
18.		
19.		
20.		
Total.....		

IX.OBJECT ASSEMBLY

	Response	Score
Manikin	120"	
Horse	180"	
Face	180"	
Auto	180"	
Total.....		

X.CODING

Time (120).....Score (No. Right)

IX. MAZES:

Maze	Maximum Errors	Errors	Score
A. 30"	2		
B. 30"	2		ss
C. 30"	2		
1. 30"	3		
2. 45"	3		
3. 60"	5		
4. 120"	6		
5. 120"	8		
Total.....			

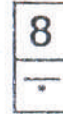
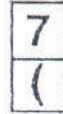
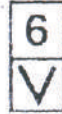
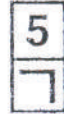
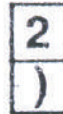
CODING A
(5-7)

SAMPLE



TIME (120*) _____ (NO. RIGHT) _____ SCORE _____

CODING B
(8-15)



SAMPLE

2	1	4	6	3	5	2	1	3	4	2	1	3	1	2	3	1	4	2	6	3	1	2	5	1

3	1	5	4	2	7	4	6	9	2	5	8	4	7	6	1	8	7	5	4	8	6	9	4	3

1	8	2	9	7	6	2	5	4	7	3	6	8	5	9	4	1	6	8	9	3	7	5	1	4

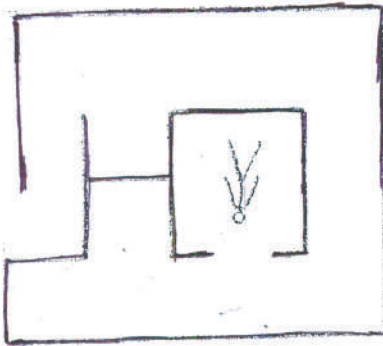
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TIME (120) _____ SCORE (NO. RIGHT) _____

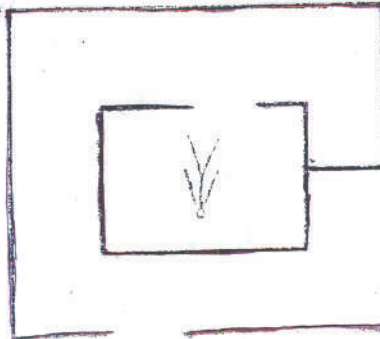
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SUBJECT

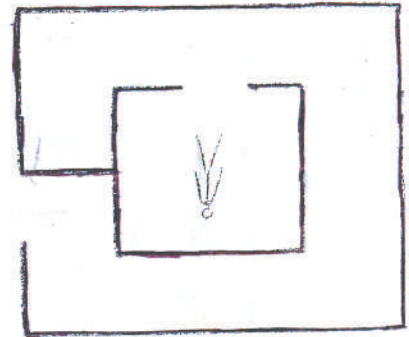
NAME.....



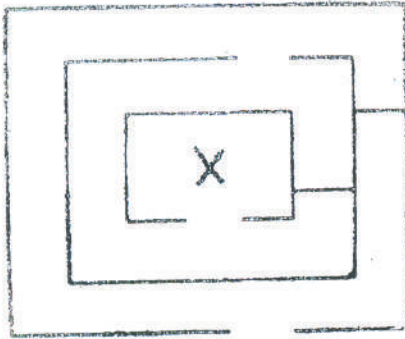
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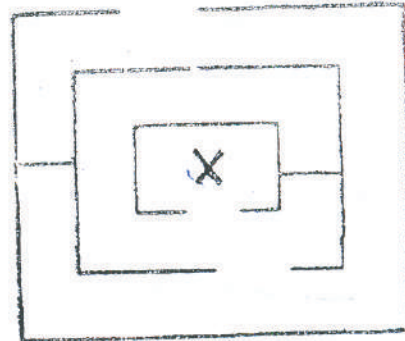
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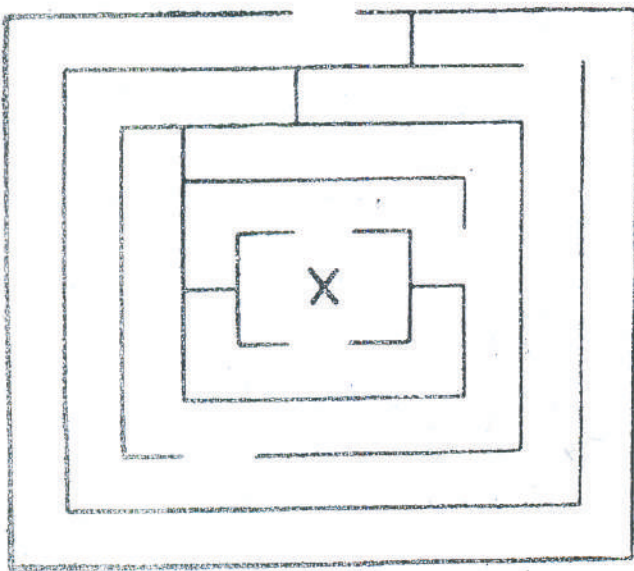
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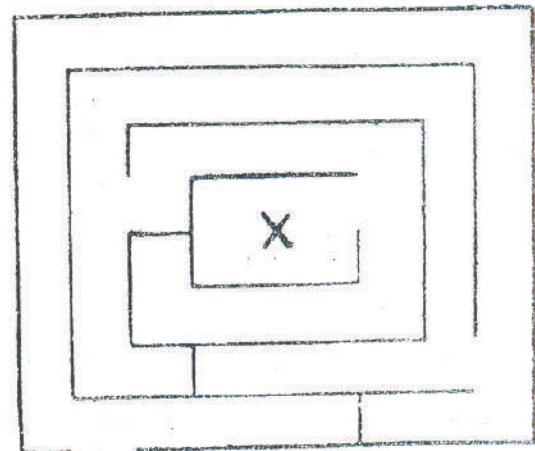
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E
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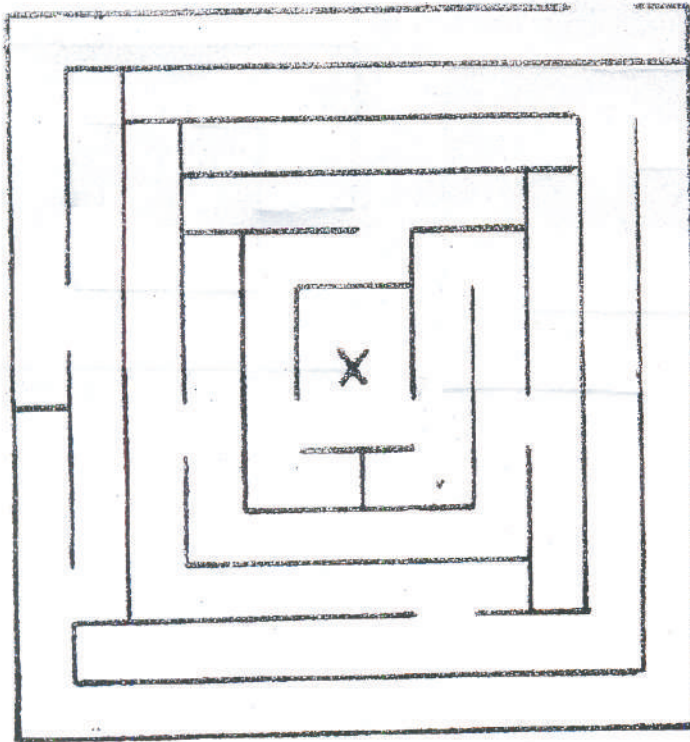
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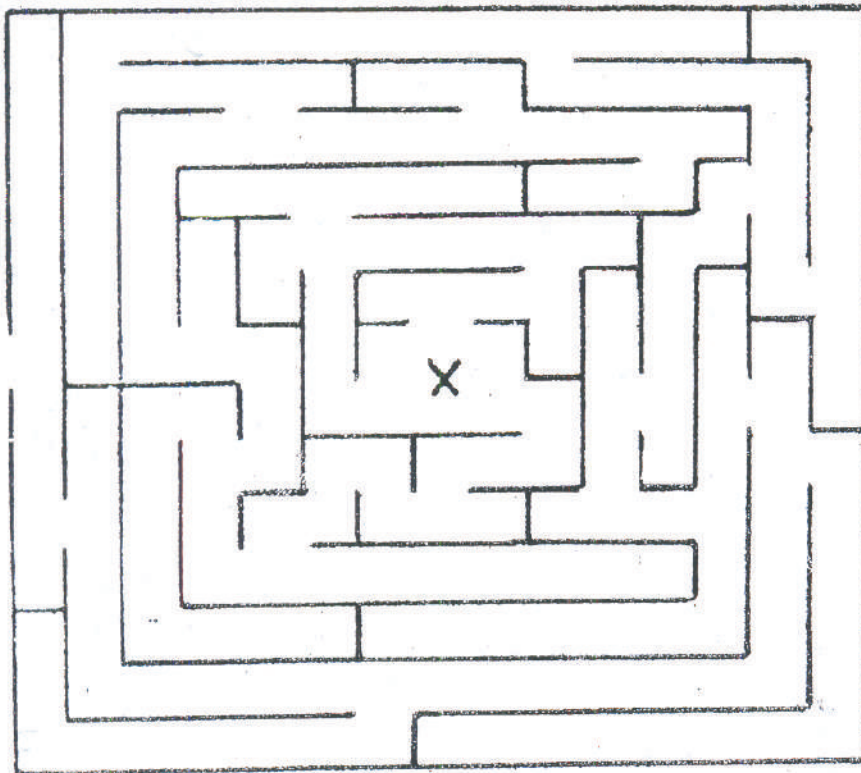
G
45

EXAMINER

SUBJECT



120



120

EXAMINER

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